

# Miaohui Shi

施妙辉

Algorithm Engineer | High-performance Computing & Robotics

Taizhou, Zhejiang, China | [shimiaohui0426.github.io](https://github.com/shimiaohui0426)

[GitHub / ShiMiaohui0426](#) • [LinkedIn / Miaohui Shi](#)

Algorithm engineer developing C++ / Python software for computational lithography, with experience in performance optimization, distributed data pipelines and equipment diagnostics. M.S. from Waseda University; research in HRI and tactile sensing.

**800 → 24 min**

**GDS end-to-end runtime**

33.3× incl. platform changes

**< 120 GB**

**Peak memory per node**

Resolved OOM on 256 GB nodes

**4 / 4**

**Subsystems with unified logs**

FastAPI • persistent task logs

## PROFESSIONAL EXPERIENCE

### Taizhou Optoelectronics Industry Innovation Center

Sep 2024-present

Algorithm Engineer

- Designed boundary-focused subpixel rasterization; ported Python to C++ and deployed across six nodes. Same GDS / 4-pass workload: 800 to 24 min (33.3× end to end, including platform changes).
- Partitioned reads: 92 to 44 min on unchanged hardware and node count. Block processing, preallocation and buffer reuse resolved OOM on 256 GB nodes; peak memory below 120 GB per node.
- Designed a shared-filesystem pipeline to compute during equipment processing, with backpressure and data-ready signals. Persisted task progress for checkpoint recovery; verified interruption/restart behavior.
- Built a FastAPI logging platform covering all four equipment subsystems, with persistent task logs and device/task queries for debugging.
- Maintain core code and operation documentation for remote support; collaborate on manual regression using a fixed GDS test library.

### Hesai Technology / Sharp-related R&D

Apr-Aug 2024

Algorithm Development

- At Hesai Technology, developed LiDAR functional-degradation algorithms and trained noise-segmentation models; also contributed to Sharp-related tactile-sensor R&D.
- Designed and trained vision-based tactile-sensor mechanical models; developed software on Ubuntu.

## EDUCATION

### Waseda University

M.S., Mechanical Engineering

2022-Mar 2024 • Japan

### Huazhong University of Science and Technology

B.S., Mechanical Science & Engineering

2017-2021 • China

## TECHNICAL SKILLS

### Performance & engineering

C++ • Python • Linux • distributed processing • memory / I/O optimization • GDS rasterization

### Robotics & interaction

ROS • Unity • HRI experiments • PASAT / NASA-TLX • behavioral analysis

### Vision & learning

YOLO • segmentation • anomaly detection • model training • vision-based tactile sensing

### Software & interfaces

FastAPI • HTTP APIs • logging • checkpoint recovery • C# / Qt • TwinCAT / PLC

## SELECTED RESEARCH

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### Multi-robot interaction, attention and automation | CASE 2024

Independently developed the experimental system (excluding the mechanical-head module), designed and conducted the experiments, and completed all data analysis. Adapted the PASAT auditory-addition task to a dual-task study involving three robotic arms, with NASA-TLX and user-experience measures.

### High-speed vision-based tactile sensing | ICAT-EGVE 2024

Independently developed algorithmic principles, implementation, performance tuning and the front-end UI. The method combines local sparse marker tracking with force estimation using an asymmetric stiffness-coefficient matrix. The paper reports 601.25 Hz sensor-system force acquisition; this is not a standalone algorithm benchmark.

## PUBLICATIONS

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- 2024      High-Speed Vision-Based Haptic Sensor for Robotic Dermatological Palpation: Force Sensing Method Using Asymmetric Stiffness Coefficient Matrix**  
Fumihiro Kato, **Miaohui Shi**, Kaito Kamishima, Hiroyasu Iwata  
ICAT-EGVE 2024
- 2024      The Role of automation and User Experience in Multi Robot Systems**  
Kouta Suzuki, **Miaohui Shi**, Vimolmongkolporn Vitvasin, Yukiko Iwasaki, Hiroyasu Iwata  
IEEE CASE 2024 • pp. 3818-3825
- 2023      Balancing Real-world Interaction and VR Immersion with AI Vision Robotic Arm**  
Xiaoran Yang, Yang Zhan, Yukiko Iwasaki, **Miaohui Shi**, Shijie Tang, Hiroyasu Iwata  
IEEE ICMA 2023 • pp. 2051-2057
- 2021      基于TwinCAT的机器人实验平台设计**  
黄弢, 施妙辉, 汪迪, 王峻峰  
实验技术与管理 • 38(12):197-202+207
- 2021      A Remote Experiment Platform based on HTML5 for Industrial Robots**  
Tao Huang, **Miaohui Shi**  
MED 2021 • pp. 920-923

## PATENTS

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### CN119987127B | Subpixel image processing based on grid-partitioned mask regions

Co-inventor | Granted 20 March 2026

Full patent record / 8 applications

## EARLIER ACADEMIC EXPERIENCE

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Oct-Dec 2022 | Learning Assistant • Waseda University

Dec 2019-Feb 2022 | Research Assistant • Huazhong University of Science and Technology

## RESEARCH DIRECTION

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Future doctoral research interest: household service robots, tactile perception, human-robot collaboration and appropriate automation. Industry interests include EDA, computational lithography and high-performance algorithm engineering.